

$$n = 2, \quad K = \mathbb{Q}(\sqrt{-2})$$

$$\tau = 1 + \frac{1}{4}\sqrt{-2}, \quad N(\tau) = \frac{9}{8}$$

$$\tau^4 = \frac{17}{64} + \frac{7}{8}\sqrt{-2}$$

$$\iota(\tau)^*|_{W(A)} = \begin{pmatrix} \frac{17}{64} & -\frac{7}{8} \\ \frac{7}{4} & \frac{17}{64} \end{pmatrix}$$

$$\iota(\tau)^*\eta^2 = N(\tau)^2\eta^2 = \frac{81}{64}\eta^2$$

$$N(a\omega_1 + b\omega_2) = a^2 + \frac{1}{2}b^2$$

